

First steps: Building and Running

Before you start to create your own programs, you need to learn how they are built and run. In this task, your aim is to learn how to setup your computer with the tools to build software, and how to use these to build and run programs from supplied code.

Key aspects include:

- Starting to see how to picture a program, using the digital reality metaphor.

Show us that you can:

- Build and run C# programs, when given the program's code:
 - Use the terminal to create a folder for the project
 - Create the project using the dotnet tool and add packages
 - Edit the code in VS Code
 - Build and run the program from the terminal in VS Code
- Build and run C/C++ programs, when given the code:
 - Create a cpp file to store the code
 - Edit the code in VS Code
 - Compile and run at the terminal in VS Code
- Capture your learning in the Learning Journey section. Include evidence (screenshots) of your progress, and your thoughts on what you accomplished, what challenges you faced, and what you found interesting.
- Provide sincere *reflections* in the Reflection section. Don't just tell us what you did, tell us what you *learned*. Reflections that are brief or shallow may not be accepted.

Complete the following:

1. Download and open the template from OnTrack.

2. Start by reviewing the steps in the [Installation overview](#) within the **V1 Programmers Guide** to get VS Code and SplashKit installed. Make sure to capture this in your learning journey.

Problems with installation?

Visit the SIT102 Teams channel where there is a Setting-Up Discussions channel – in there is a handy video to watch and a troubleshooting guide.

3. Follow the guided tour in [Chapter 3 Building Programs](#) – make sure to add this to your learning journey.
4. You should have enough to submit... but if you want to explore this further, skim over the first few chapters for context, and to see more about using the terminal.
 - a. [Part 0: First Steps](#) – get an idea of how the material is organised.
 - b. [Part 0 - Chapter 1: Digital Realities](#) – think about what programs are.
 - c. [Part 0 - Chapter 2: Computer Use](#) – explore terminal commands in depth.
 - d. [Part 0 - Chapter 3: Building Programs](#) – read about what is happening with compiling and running.

Feeling a bit lost?

No problem at all! Your tutors are experts and would love to assist. Please reach out during your seminars (you can attend as many seminars as you like) or write a message on the SIT102 Teams channel.

5. Complete your submission – complete the summary and related activities in the template.
6. **Upload** to OnTrack and **engage with feedback** to get this complete!

Available Resources

We have many resources to assist in your learning journey – if you are ever unsure then please reach out to us, we are always happy to help.

- **The SIT102 Teams channel** – an active learning resource with weekly announcements from the Unit Chair, and regular discussions/assistance from the SIT102 teaching team and students.
- **Seminars** – you can attend both on-campus and online sessions, and join as many as you like each week. Your tutors will provide demonstrations and assistance to help guide your learning.
- **HelpHubs** – there are sessions hosted nearly everyday, both on-campus and online providing one-on-one assistance from a tutor. Please join as many as you like.
- **PASS sessions** – running every week, these sessions are run like a study group, hosted by a past SIT102 student.

For more information about the above resources please speak to your tutor or visit the SIT102 Teams channel.

Caution: Acceptable use of Artificial Intelligence

It is acceptable to use AI to improve your style, as you would be using a spell checker to improve your spelling. Do **not** ask an AI to write the summary for you: this is cheating because it does not demonstrate that you understand anything.